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Started	Friday, 30 October 2026, 10:00 PM
Completed	Friday, 30 October 2026, 10:06 PM
Duration	6 mins 24 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 10 ✓
b. 1000
c. 100
d. 500

Question **2**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By comparing each element sequentially
b. By starting from the first element
c. By using a hash function
d. By dividing the list length by two ✓

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It can only be applied to large lists
b. It is slower than linear search for small lists
c. It does not work with negative numbers
d. It requires the list to be sorted ✓

Question **4**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Binary search
b. Jump search
c. Hashing
d. Linear search ✓

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Binary search algorithm is not efficient when the data elements more than 1500 ✓
b. Must use a sorted array
c. Requirement of sorted array is expensive when a lot of insertion and deletions are needed
d. There must be a mechanism to access middle element directly

Question **6**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search starts over from the beginning
b. The search stops immediately ✓
c. The search backtracks to find duplicate elements
d. The search continues until the end of the list

Question **7**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search stops
b. The search continues in the left or right sublist ✓
c. The search continues from the beginning
d. The list is sorted

Question **8**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(1)
b. O(n log n)
c. O(log n) ✓
d. O👎

Question **9**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. log n
b. n ✓
c. 1
d. n/2

Question **10**

Incorrect

Mark 0.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

a. Binary search
b. Interpolation search
c. Hash search ✗
d. Linear search

Question **11**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the right sublist
b. In the left sublist ✓
c. In the entire list
d. At the beginning of the list

Question **12**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 90 and 100
b. 89 and 94
c. 94 and 99
d. 90 and 99 ✓

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Dividing the list in half repeatedly
b. Skipping every second element
c. Checking each element sequentially ✓
d. Sorting the list before searching

Question **14**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O👎 ✓
b. O(n log n)
c. O(log n)
d. O(1)

Question **15**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

a. Bubble search
b. Merge search
c. Quick search
d. Linear search ✓